Behavioural Profiles Are Not Heritable Sequences: a Refuted Transfer of Phylogenetic Recombination Detection to Genetic Programming

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Abstract

Evolutionary program synthesis in the AlphaEvolve family applies mutation and selection to populations of programs, usually with a crossover operator whose contribution is asserted rather than measured. Phylogenetics has an exact instrument for precisely that question: a quartet certificate refutes the four-point condition on four taxa with an integer comparison a proof kernel can recheck, and a recombinant genealogy violates that condition where a clonal one does not. This paper proposed transferring the instrument, tested the transfer, and reports its failure together with the reason.

The transfer looked clean. A candidate program's *behavioural profile* — its outcome at each of T probes, drawn from {correct, incorrect, raised, non-finite} with a gap symbol for an inadmissible probe — is a four-state sequence with missing data, which is exactly the model a nucleotide alignment assumes. A programs × probes matrix is therefore an alignment, and the certificate machinery runs on it unmodified. A natural null comes for free: a population varied only by mutation from single parents is a tree *by construction*.

On synthetic profiles the transfer appeared to work, separating clonal from recombinant arms by margin across a usable window of mutation rates. On real program populations — expression trees over a closed grammar, evolved against a rational target, every individual executed at 600 probes — it fails. Three pre-registered controls fail in the same direction: the recombinant arm exceeds the clonal floor in 0 of 24 runs, 0 of 23 when restricted to comparable best fitness, and 0 of 22 when restricted to comparable *realised behavioural spread*. Stratifying all 49 populations by spread then explains everything: above a spread of 0.05, 4 of 4 clonal populations carry certificates and so does 1 of 1 recombinant; below 0.01, none of 29 does in either arm. Behavioural spread accounts for certificate presence and arm membership adds nothing.

The diagnosis is specific and it indicts the proposal rather than the implementation. The four-point condition detects non-tree-like structure in a distance matrix; it detects *recombination* only when the characters are independently heritable units, because recombination's signature is mosaic ancestry *along* the sequence. A probe outcome is not inherited: it is a consequence of the whole program, so two unrelated point mutations routinely alter overlapping sets of probe outcomes and manufacture the same pattern. The synthetic validation concealed this because its profiles were generated by mutating profile *strings*, which builds site-wise heritability in by construction — it confirmed that the method detects recombination in site-wise-heritable data, which was never in doubt.

We report what survives: the proof-carrying fitness ladder, which is independent of this result; the twelve measurement guards, three of which caught this failure; and a released dataset of 49 evolved program populations with behavioural profiles, certificates and fitness traces.

1 What was proposed, and why it was plausible

Systems in the AlphaEvolve family evolve programs with an LLM or with syntax-level operators, score them against an evaluator, and keep what scores better. Most carry a crossover operator. Whether crossover contributes is an old and unsettled question in genetic programming, and in practice it is answered by assertion: the operator is present, so recombination is assumed to occur.

Phylogenetics answers the analogous question exactly. The four-point condition [1] holds on every tree metric,

$$d(a,b) + d(c,e) \;\le\; \max\big(d(a,c) + d(b,e),\; d(a,e) + d(b,c)\big), \tag{1}$$

and a *quartet certificate* is four taxa whose six pairwise counts, over the one denominator they share under complete deletion within the quadruple, refute (1) as an integer comparison. Prior work in this programme built such certificates, proved their relationship to split decomposition [2], and emitted them for kernel rechecking [3, 4].

The transfer is immediate once one notices the following.

Definition 1.1 (Behavioural profile)**.** Run a candidate program against T fixed probes and record one symbol per probe: `A` correct, `C` finite but incorrect, `G` raised, `T` non-finite, and `-` where the probe is inadmissible.

That is a four-state alphabet with a gap symbol, which is the model a nucleotide alignment assumes; the gap is handled by the same complete deletion that handles a missing base. A programs × probes matrix is thus a valid alignment and the machinery runs on it with no modification.

The null is better than convenient, it is correct: a population whose only variation is mutation from single parents has a genealogy that is a tree *by construction*. So the question "does this crossover operator recombine?" becomes "does the crossover-enabled population's margin exceed the floor of a mutation-only population at matched budget?" — with an exhibit a kernel can recheck.

2 The synthetic test, which appeared to confirm it

Eight candidates, 1200 probes, three seeds per cell, two arms at matched budget: a mutation-only lineage, and the same lineage with crossovers at interior breakpoints. Mutation rate swept, because a statistic of this kind dies at both ends.

rate	clonal violating	recomb. violating	max clon. margin	min rec. margin	margin sep.	null varies
0.0005	[0, 0, 0]	[0, 0, 0]	0	0	no	**no**
0.0020	[0, 0, 0]	[21, 25, 27]	0	0.00342	yes	**no**
0.0080	[13, 19, 21]	[37, 53, 45]	0.00342	0.00514	**yes**	**yes**
0.0200	[60, 61, 56]	[59, 59, 65]	0.00685	0.01370	**yes**	**yes**

Three readings, each a known lesson of this programme reappearing. Certificates are found and all revalidate. The *margin* is the statistic and the count is not: at rate 0.02 the count fails outright (clonal 56–61 against recombinant 59–65) while the margin still separates twofold. And both extremes are unusable for opposite reasons — below 0.002 the clonal null produces no violating quadruple at all, so any positive observation is trivially significant and must be reported as degenerate rather than as a result.

This is the experiment that should have been distrusted, and §4 says why.

3 The real test

Representation. Expression trees over a closed grammar: terminals x and small integer constants, operators $+$, $-$, $\times$ and a protected division that *raises* on a zero denominator rather than returning a fallback, so a structural failure stays distinguishable from a wrong answer. The grammar cannot express an import, an attribute access or a call, so individuals are safe to evaluate directly and no sandbox is load-bearing.

Task and probes. Target $(x^3 - 2x + 1)/(x^2 + 1)$, which makes division genuinely necessary so the `G` state is reachable. 600 probes drawn uniformly on $[-2, 2]$; tolerance 0.01.

Arms. Both spend the same number of variation events per generation and cap program size, so neither event budget nor bloat can confound. The clonal arm applies point mutations only — a tree by construction. The recombinant arm routes a fraction of its events through subtree exchange, which is a recombination with a breakpoint; that fraction is swept over $\{0.1, 0.2, 0.4, 0.6\}$.

Pre-registered before any run.

P1 the clonal null is non-degenerate: it produces violating quadruples, so something can be measured against it;

P2 the floor converges: 25 clonal runs do not raise the maximum above the first 5;

P3 the recombinant margin exceeds the clonal floor, decided on the integer count first and only then as a rate;

P4 the effect is not an artifact of fitness: it survives restricting both arms to comparable best fitness;

P5 the effect is not an artifact of divergence: it survives restricting the recombinant arm to runs whose *realised behavioural spread* lies inside the clonal arm's range.

Result. P1 holds: 5 of 25 clonal runs carry certificates, with up to 30 violating quadruples. P2 holds exactly: the floor at 5 runs and at 25 runs are both 0.015000, a growth factor of 1.00.

pre-registered prediction	held	of
P3 recombinant margin exceeds the clonal floor	**0**	24
P4 ... restricted to comparable best fitness	**0**	23
P5 ... restricted to comparable realised spread	**0**	22

23 of 24 recombinant runs produce *zero* violating quadruples, at every crossover share. P5 matters most: 22 of the 24 were inside the clonal spread range, so the failure is not a matching artifact. Crossover also bought no fitness — clonal median best error 0.5707 against recombinant 0.6028, with the best individual overall identical in both arms at 0.4745.

What actually drives the certificates. Pooling all 49 populations and stratifying by realised behavioural spread:

spread	arm	runs	with certificates	mean violating
$[0, 0.01)$	clonal	12	0	0.0
$[0, 0.01)$	recombinant	17	0	0.0
$[0.01, 0.05)$	clonal	9	1	1.2
$[0.01, 0.05)$	recombinant	6	0	0.0
$[0.15, 1.00)$	clonal	4	**4**	21.8
$[0.15, 1.00)$	recombinant	1	**1**	9.0

Above a spread of 0.05 every population carries a certificate in both arms; below 0.01 none of 29 does in either. **Behavioural spread accounts for certificate presence and arm membership adds nothing** — and the clonal arm, which cannot recombine, carries them at the same rate.

4 Diagnosis: heritability is the load-bearing assumption

The four-point condition detects non-tree-like structure *in a distance matrix*. It detects *recombination* only under an additional assumption that nucleotide data satisfies and behavioural profiles do not.

Remark 4.1 (Why recombination is visible in DNA). A nucleotide site is an independently heritable unit: a child's state at site i is its parent's state at site i, modified rarely and locally. Recombination produces *mosaic ancestry along the sequence* — a prefix from one parent and a suffix from another — so different sites carry different genealogies and the resulting distances cannot be embedded in one tree. That conflict is what (1) detects.

A probe outcome has no such property. It is not inherited; it is *computed* from the whole program. A single point mutation near the root of an expression tree changes the outcome at many probes at once, and two unrelated mutations in two lineages routinely change overlapping probe sets. That manufactures exactly the pattern of conflicting distances that (1) reports, with no recombination anywhere — which is precisely what the clonal arm's 4/4 at high spread shows.

The synthetic validation was circular. Its profiles were produced by mutating profile *strings* position by position, which *builds site-wise heritability in by construction*. It therefore confirmed that the method detects recombination in site-wise-heritable data — which was never in doubt — while appearing to validate an application in which that property fails. This is a non-discriminating check of the kind this programme has had to label before: a test both hypotheses pass. The distinguishing experiment is the one in §3, where profiles are *measured* from executed programs instead of generated, and it reverses the conclusion.

5 What survives

The proof-carrying fitness ladder is untouched. The second half of the proposal replaced an empirical objective with one whose upper rungs are Lean 4 obligations read through `#print axioms`, so that no amount of latency gain buys a rung and `sorry` cannot pass. Nothing here bears on it, and the motivation stands: an un-kernel-checked gate is eventually found to be forced, degenerate, or measuring the apparatus.

Three guards earned their place. The programme's measurement rules caught this rather than ratifying it: the demand that a null be able to vary flagged the degenerate low-rate cells; the demand that a floor be shown converged is what made 0/24 trustworthy; and the demand to match on the *realised* nuisance rather than the nominal budget produced P5, without which the failure would plausibly have been dismissed as a spread artifact.

A released dataset. 49 evolved program populations — 25 clonal and 24 recombinant across four crossover shares — each with its behavioural profile matrix in FASTA, violating-quadruple counts, exact margins, realised spread, distinct-profile counts, best fitness and the best program rendered as an expression. Useful to anyone asking whether a crossover operator contributes, independently of the method refuted here.

A transferable negative. Before carrying a phylogenetic method to a non-biological population, ask what the method assumes about *heritability of characters*, not merely about the alphabet. Matching the data model — four states and a gap — was necessary and nowhere near sufficient.

6 Limitations

One task, one grammar, one population size, and tournament selection with elitism; subtree exchange is one crossover among several. A different representation with genuinely modular, independently heritable components — separate functions recombined as units, say — might restore the assumption §4 identifies as missing, and that is the obvious next experiment rather than a defence of this one. The negative result is about the transfer as proposed and tested, not a theorem about all program populations.

The fitness comparison is also underpowered for its own question: 25 against 24 runs on one task cannot settle whether crossover helps GP in general, and no such claim is made.

References

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